Product Quality Standard of FR-4/G10/FR-5/G11 Epoxy Glass Fiber Reinforced Laminate

I. TECHNICAL REQUIREMENTS

- surface

1. The product surface should be flat, smooth, and free of obvious scratches, wrinkles, or cracks.
2. The product should be clean, free from oil contamination, and devoid of visible impurities.
3. The color of the same batch and the same product should remain essentially consistent, with minor white edges allowed as long as they do not affect usability, but no two or more distinct color differences should be present.
4. The product must not be damaged or have any missing corners.
5. The product cross-section must be cut neatly with no cracks.
6. The product must not show any layering;

- dimensional tolerance

The allowable tolerances for product length and width shall comply with the specifications in Table 1.

Table 1 (mm)

Width/Length	Length tolerance
510～1000	±15
＞1000	±25

- thickness deviation

(1) The nominal thickness and its tolerance of the product shall comply with the specifications in Table 2.

Table 2-A: Product Nominal Thickness and Tolerance Specifications (mm)

Nominal thick-ness	Common difference	Nominal thick-ness	Common difference	Nominal thick-ness	Common difference
0.100	±0.0180	1.600	±0.1300	15.875	±1.0160
0.150	±0.0250	2.000	±0.1800	19.050	±1.0922
0.200	±0.0380	2.500	±0.1800	22.225	±1.1684
0.250	±0.0380	3.000	±0.2300	25.400	±1.2446
0.300	±0.0500	4.000	±0.3000	28.575	±1.3462
0.400	±0.0500	4.500	±0.3000	31.750	±1.3970
0.500	±0.0640	5.000	±0.3000	34.925	±1.4732
0.600	±0.0640	6.000	±0.3000	38.100	±1.5494
0.800	±0.1000	8.000	±0.3000	41.275	±1.6256
1.000	±0.1000	10.000	±0.3000	44.450	±1.7018
1.200	±0.1300	11.125	±0.8382	47.625	±1.7780
1.500	±0.1300	12.700	±0.9144	50.800	±1.8542
Remarks	1: For products with non-standard thickness, the manufacturer and customer may negotiate the production, with deviations calculated using the tolerance of the next larger standard thickness.				

Nominal thick-ness	Common difference	Nominal thick-ness	Common difference	Nominal thick-ness	Common difference
0.40	±0.10	3.00	±0.37	20.00	±1.30
0.50	±0.12	4.00	±0.45	25.00	±1.50
0.60	±0.13	5.00	±0.52	30.00	±1.70

0.80	±0.16	6.00	±0.60	35.00	±1.95
1.00	±0.18	8.00	±0.72	40.00	±2.10
1.20	±0.20	10.00	±0.82	45.00	±2.30
1.60	±0.24	12.00	±0.94	50.00	±2.45
2.00	±0.28	14.00	±1.02	/	/
2.50	±0.33	16.00	±1.12	/	/
Remarks	1 For products with non-standard thickness, the manufacturer and customer may negotiate the production, with the deviation being the difference between the standard thickness and the next higher standard thickness.				

- angularity

The warpage of the product shall comply with the requirements specified in Table 4.

Table 4: Product Warpage Requirement Table (mm)

Thickness	Angularity	
	1000 (ruler length)	500(Ruler length)
3.0～6.0	≤10	≤2.5
6.1～8.0	≤8	≤2.0
＞8.0	≤6	≤1.5

- machining

After mechanical processing such as sawing, drilling, turning, and milling on FR-4/G10/FR-5/G11 epoxy boards, no cracks, delamination, or slagging should occur.

- The product technical parameters shall comply with the specifications in Table 5.

Table 5: Technical Parameters of Epoxy Board (FR-4/G10/FR-5/G11)

Identification of product				E3501	E3502	3500	HT-E3500
International Electrotechnical Commission/ British Standard				EPGC 201	EPGC 202	EPGC 203	EPGC 204
American (International Electrical) Standard				G10	FR-4	G11	FR-5
German standard				Hgw2372.0	Hgw2372.1	Hgw2372.4	Hgw2372.2
Japanese standard				-	KEL-GEF	-	-
Project		Test method	Unit	Numeric value	Numeric value	Numeric value	Numeric value
Physical property	Pigment	—	—	Green/Yellow			
	Proportion	—	g/cm^3	1.95 - 2.0	1.95 - 2.0	1.95 - 2.0	1.95 - 2.0
	Water absorption	E-24/50+D-24/23	%	0.07-0.16	0.07-0.16	0.07-0.16	0.07-0.16
Behaviour of electricity	Electrical strength (vertical)	C-96/20/65(in 90℃ Oil)	KV/mm	≥ 14	≥ 14	≥14	≥ 14
	Breakdown voltage (parallel)	C-0.5/90 (in 90℃ Oil)	KV	40	40	40	40
	Mass resistivity	C-96/20/65	Ω	5×10^{11}-10^{13}	5×10^{11}-10^{13}	5×10^{11}-10^{13}	5×10^{11}-10^{13}
		C-96/20/65+C-96/40/90		5×10^{10}-10^{12}	5×10^{10}-10^{12}	5×10^{10}-10^{12}	5×10^{10}-10^{12}
	Surface resistivity	C-96/20/65	Ω	5×10^{10}-10^{12}	5×10^{10}-10^{12}	5×10^{10}-10^{12}	5×10^{10}-10^{12}
		C-96/20/65+C-96/40/90		5×10^{10}-10^{11}	5×10^{10}-10^{11}	5×10^{10}-10^{11}	5×10^{10}-10^{11}
	Insulation strength	C-96/20/65	Ω--cm	5×10^{11}-10^{12}	5×10^{11}-10^{12}	5×10^{11}-10^{12}	5×10^{11}-10^{12}
		C-96/20/65 +D-2/100		5×10^{8}-10^{9}	5×10^{8}-10^{9}	5×10^{8}-10^{9}	5×10^{9}-10^{11}
	Dielectric constant (1 MHz)	C-96-20-65	---	4.0-5.0	4.0-5.0	4.0-5.0	4.0-5.0
		C-96/20/65 +D-48/50		4.5-5.5	4.5-5.5	4.5-5.5	4.5-5.5

	Medium dissipation factor (1 MHz)	C-96/20/65	---	0.03-0.04	0.03-0.04	0.03-0.04	0.03-0.04
		C-96/20/65+D-48/50		0.04-0.05	0.04-0.05	0.04-0.05	0.04-0.05
	Arc resistance (high voltage)	C-96/20/65	Sec	130-140	130-140	130-140	130-140
Mechanical behavior	Flexural strength (transverse)	A	kg/mm^2	40-50	40-50	40-50	40-50
	Impact strength (parallel)	A	KJ/m^2	33	33	33	33
	Rockwell hardness	A	M	110	110	110	110
	Cohesional strength	A	N	8000	8000	8000	8000
	Compress Intensity Perpendicular	A	kg/mm^2	29-34	29-34	29-34	29-34
	Compress Intensity Parallel			19-24	19-24	19-24	19-24
Combustibility	UL inflaming retarding	A	---	94 HB	94V-O	94 HB	94V-O
	Heat-resisting	A (120min)	℃	220	220	220/240/280	220/240/280
	Continuous operating temperature	---	℃	160	160	160/180/200	160/180/200
	Smoke density	ASTM-E 662	—	1	1	1	1
	Smoke density			3	3	3	3
	Toxin content	—		Low	Low	Low	Low
Other	Antipropion	boiled	Min	30(OK)	30(OK)	30(OK)	30(OK)
	Stamping property	A	℃	---	---	---	---
Remarks	The above data represent typical values for thicknesses of 1.5mm or 1.6 mm.						

II: Product Inspection Methods

- Appearance: Visual inspection.
- Dimensional tolerance: Use a 1mm ruler or tape measure to measure three points along the width or length of the board, and take the average value respectively.
- Thickness tolerance: Measure thickness at 2 points (total 8 points) with micrometer/vocabular caliper at 20mm from each edge of the plate, with precision to 0.01mm. The test result is the median of 8 measurements, and the maximum and minimum values are reported.
- Torsion: Place the sheet to be measured with its concave side up on the platform, then position a lightweight straightedge (1000mm or 500mm, weighing less than 500g) on the sheet. Measure the maximum distance between the upper surface of the sheet and the straightedge, with the result expressed in mm.
- Other items were tested according to GB-T 5130-1997 Test method for industrial rigid laminates made of thermosetting resin for electrical use.

III. Product Inspection, Storage, and Transportation

- Inspection: Appearance, dimensional tolerances, and thickness tolerances are mandatory routine inspection items for the product. Other items are inspected only upon customer request.
- Storage: The laminated board shall be placed in a dry and clean room with a temperature not exceeding 40°C, on a pad no less than 5cm above the ground, and shall be laid flat. It must not be placed near heat sources, heating systems, or direct sunlight.
- Transportation: During transportation of laminated materials, mechanical damage, moisture exposure, and sunlight exposure should be prevented.

IV. APPLICATION TIPS

- FR-4/G10/FR-5/G11 epoxy laminates have low thermal conductivity, so machining should be performed at high cutting speeds and small feed rates.
- A large amount of dust and odor are produced during machining, so it is necessary to strengthen labor protection.
- The method JB/Z141-1979 is adopted in the machining to avoid loss.